MODULE 5 BRAIN

The Brain.

THE CENTRAL NERVOUS SYSTEM, THE BRAIN

Covers the development of the brain, the cerebrum with its lobes, white matter tracts, and basal ganglia, the functional areas of the cortex, the diencephalon, the cerebellum, the limbic system, and the arterial supply of the brain. The focus is on the structures and the job each one does.

BY THE END

1. Trace the development of the brain from the neural tube vesicles.
2. Identify the cerebrum, its lobes, white matter tracts, and basal ganglia.
3. Map the functional areas of the cerebral cortex.
4. Name the parts of the diencephalon, the cerebellum, the limbic system, and the arteries that supply the brain.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brain

drsrennie-stack.github.io/new-build-bio4-solano/brain-concept-videos.html

The Brain, an Overview

The brain sits within the cranial cavity and holds roughly 85 billion neurons. It is the body's master control and integration center. The brain has four major parts.

THE FOUR MAJOR PARTS, AND THE TWO KINDS OF NERVOUS TISSUE

Structure	What it is
Brain	The organ within the cranial cavity, the control and integration center of the body
Cerebrum	The largest brain part, the seat of conscious thought, sensation, and voluntary movement
Diencephalon	The deep brain region between the cerebrum and the brainstem
Brainstem	The stalk connecting the brain to the spinal cord, covered on its own sheet
Cerebellum	The structure behind the brainstem that coordinates movement
Gray matter	Neuron cell bodies, dendrites, and unmyelinated fibers; on the outside of the brain
White matter	Bundles of myelinated axons; on the inside of the brain

Development of the Brain

The brain and spinal cord develop from the ectodermal neural tube. The tube swells at its anterior end into a series of vesicles that mature into the adult brain.

FROM TUBE TO VESICLES

Structure	What it is
Neural tube	The ectodermal tube from which the entire central nervous system develops
Primary brain vesicles	The first three swellings: the prosencephalon (forebrain), the mesencephalon (midbrain), and the rhombencephalon (hindbrain)
Secondary brain vesicles	The prosencephalon divides into the telencephalon and diencephalon; the rhombencephalon divides into the metencephalon and myelencephalon

Each secondary vesicle matures into specific adult structures. Compare them.

SECONDARY VESICLES AND WHAT EACH BECOMES

Secondary vesicle	Adult structures
Telencephalon	The cerebrum and the lateral ventricles
Diencephalon	The thalamus, hypothalamus, epithalamus, and third ventricle
Mesencephalon	The midbrain and the cerebral aqueduct
Metencephalon	The pons, the cerebellum, and the upper fourth ventricle
Myelencephalon	The medulla oblongata and the lower fourth ventricle

KEY POINT

The secondary vesicle list is the adult brain in order, from front to back: cerebrum, diencephalon, midbrain, pons and cerebellum, medulla. Learn the five vesicles and you have already named the major regions.

The Cerebrum

The cerebrum is the largest region of the brain. Its surface is folded to pack more cortex into the skull, and a thick core of white matter lies beneath.

THE SURFACE AND THE CORE

Structure	What it is
Cerebral hemispheres	The right and left halves of the cerebrum, joined by the corpus callosum, each with five lobes
Cerebral cortex	The outer layer of gray matter, 2 to 4 mm thick, holding billions of neuron cell bodies
Cerebral white matter	The myelinated axons beneath the cortex that connect cortical areas to each other and to lower regions
Gyri	The raised ridges of the folded cortex
Sulci	The shallow grooves between the gyri
Fissures	The deepest grooves, which divide major regions of the brain

Specific named sulci and fissures mark the boundaries on the cerebral surface. Compare them.

THE NAMED SURFACE BOUNDARIES

Structure	What it separates
Central sulcus	The frontal lobe from the parietal lobe; it divides the primary motor cortex in front from the primary somatosensory cortex behind
Lateral sulcus	The frontal and parietal lobes from the temporal lobe
Parieto-occipital sulcus	The parietal lobe from the occipital lobe
Longitudinal fissure	The left and right cerebral hemispheres
Transverse fissure	The cerebrum from the cerebellum

The Cerebral Lobes

Each hemisphere has five lobes. Four are named for the cranial bone above them; the fifth, the insula, lies hidden deep in the lateral sulcus. Compare them.

THE FIVE LOBES, WHERE EACH SITS AND WHAT IT DOES

Lobe	Location	Primary roles
Frontal lobe	The front of the cerebrum	Voluntary movement at the precentral gyrus, motor planning, speech production at Broca's area, judgment, and personality
Parietal lobe	Behind the central sulcus	Somatic sensation at the postcentral gyrus and spatial orientation
Temporal lobe	The lower side of the cerebrum	Hearing, language comprehension at Wernicke's area, and memory
Occipital lobe	The back of the cerebrum	Visual processing
Insula	Deep within the lateral sulcus	Taste, visceral sensation, and emotion

Cerebral White Matter Tracts

The white matter beneath the cortex is sorted into three classes of fiber, grouped by where each tract begins and ends. Compare them.

THE THREE FIBER CLASSES

Fiber class	What it connects	Examples
Association fibers	Gyri within the same hemisphere, allowing communication across one side	The arcuate fasciculus, which links Broca's and Wernicke's areas
Commissural fibers	Matching gyri between the two hemispheres	The corpus callosum, the largest; also the anterior and posterior commissures
Projection fibers	The cortex with lower brain regions and the spinal cord, as ascending or descending tracts	The internal capsule, the major projection pathway beside the basal ganglia

The Basal Ganglia

The basal ganglia, also called the cerebral nuclei, are deep masses of gray matter buried within the cerebral white matter. They help regulate movement.

THE NUCLEI AND WHAT THEY DO

Structure	What it is
Basal ganglia	The cerebral nuclei, deep gray matter masses that control semi-voluntary movement, such as the arm swing of walking
Caudate nucleus	A long, curved nucleus, one of the core components
Putamen	A core component lying lateral to the globus pallidus
Globus pallidus	A core component lying medial to the putamen
Substantia nigra	A midbrain nucleus grouped with the basal ganglia by function
Subthalamic nucleus	A diencephalic nucleus grouped with the basal ganglia by function
Roles	Initiating and regulating movement, suppressing unwanted movement, and modulating muscle tone

Functional Areas of the Cerebral Cortex

The cortex is mapped into specialized areas of three kinds: motor areas that issue commands, sensory areas that receive input, and association areas that interpret and integrate.

THE THREE KINDS OF AREA

Structure	What it is
Motor areas	Issue the commands for voluntary movement; found mainly in the frontal lobe
Sensory areas	Receive and begin to process input from the body's receptors
Association areas	Integrate sensory input with memory and reasoning to plan a response

Each named area has a fixed location and a specific job. Compare them.

THE NAMED FUNCTIONAL AREAS

Area	Location	Role
Primary motor cortex	Precentral gyrus of the frontal lobe	Initiates voluntary movement
Premotor cortex	Anterior to the primary motor cortex	Plans learned, patterned movements
Supplementary motor area	Medial frontal lobe	Coordinates complex bilateral movements
Frontal eye field	Middle frontal gyrus	Controls voluntary eye movement
Broca's area	Inferior frontal gyrus, usually the left	Controls the motor production of speech
Primary somatosensory cortex	Postcentral gyrus of the parietal lobe	Receives conscious touch, pressure, and proprioception
Primary visual cortex	Occipital lobe, at the calcarine sulcus	Receives raw visual input
Primary auditory cortex	Superior temporal gyrus	Receives auditory input
Primary gustatory cortex	Insula and inferior postcentral gyrus	Receives taste input
Primary olfactory cortex	Medial temporal lobe, at the uncus	Receives smell input
Vestibular cortex	Posterior insula and the parietal cortex	Receives input for balance and spatial orientation
Prefrontal cortex	Anterior frontal lobe	Executive function: judgment, planning, and personality
Wernicke's area	The temporal and parietal junction, usually the left	Comprehension of spoken and written language
Somatosensory association area	Posterior to the postcentral gyrus	Interprets texture, shape, and spatial orientation
Visual association area	Surrounding the primary visual cortex	Recognizes faces and objects
Auditory association area	Adjacent to the primary auditory cortex	Interprets speech and music

The Diencephalon

The diencephalon links the cerebral hemispheres to the brainstem and surrounds the third ventricle. It has three parts: the thalamus, the hypothalamus, and the epithalamus.

THE THREE PARTS AND WHAT LIES WITHIN THEM

Structure	What it is
Thalamus	Paired oval masses of gray matter that form about 80 percent of the diencephalon; the main sensory relay station to the cortex
Massa intermedia	The interthalamic adhesion, a bridge joining the right and left thalami
Thalamic nuclei	The individual nuclei of the thalamus, each one relaying a specific sensory or motor modality
Hypothalamus	The region below the thalamus; the master control center for homeostasis
Hypothalamic roles	Controlling the autonomic nervous system, endocrine function through the pituitary, body temperature, hunger and thirst, and sleep-wake rhythms
Epithalamus	The small posterior part of the diencephalon
Pineal gland	The endocrine gland of the epithalamus that secretes melatonin
Habenular nuclei	Nuclei of the epithalamus involved in emotional and visceral responses to smell
Third ventricle	The midline fluid-filled cavity enclosed by the diencephalon

The Cerebellum

The cerebellum is the second largest brain region, sitting in the posterior cranial fossa below the occipital lobes. It coordinates movement, posture, and balance.

THE PARTS OF THE CEREBELLUM

Structure	What it is
Cerebellar hemispheres	The two lateral halves of the cerebellum
Vermis	The narrow midline band connecting the two hemispheres
Cerebellar lobes	Each hemisphere has an anterior, a posterior, and a flocculonodular lobe
Folia	The fine, leaf-like folds of the cerebellar surface
Cerebellar cortex	The outer gray matter of the cerebellum
Arbor vitae	The branching, tree-like pattern of cerebellar white matter
Deep cerebellar nuclei	Gray matter nuclei embedded in the white matter, such as the dentate nucleus

Three pairs of cerebellar peduncles, stalks of white matter, connect the cerebellum to the brainstem. Compare them.

THE THREE PAIRS OF PEDUNCLES

Peduncle	What it connects
Superior cerebellar peduncle	Connects to the midbrain; carries mostly output from the cerebellum
Middle cerebellar peduncle	Connects to the pons; carries input from the pontine nuclei
Inferior cerebellar peduncle	Connects to the medulla; carries input from the spinal cord and vestibular system

The Limbic System

The limbic system is a functional system, not a single structure. Its parts ring the diencephalon and reach into the cerebrum. It sets emotional state and supports memory.

THE LIMBIC STRUCTURES AND THEIR ROLES

Structure	What it is
Limbic system	A functional system spanning the cerebrum and diencephalon that sets emotional state and supports memory
Amygdala	The limbic structure for fear and aggression, linking emotion to memory
Hippocampus	The limbic structure essential for forming new long-term memories
Cingulate gyrus	The limbic gyrus arching above the corpus callosum, involved in expressing emotion
Limbic roles	Establishing emotional states and drives, linking conscious thought to brainstem responses, and supporting long-term memory

Cerebral Arterial Supply: The Circle of Willis

The brain is fed by two arterial systems that join in a ring at the base of the brain, the cerebral arterial circle, also called the Circle of Willis. The ring lets blood cross from one source to another if a vessel narrows or is blocked.

THE TWO CIRCULATIONS AND THE RING THAT JOINS THEM

Structure	What it is
Anterior circulation	The blood supply from the internal carotid arteries; it feeds most of the cerebrum
Posterior circulation	The blood supply from the vertebral and basilar arteries; it feeds the brainstem, cerebellum, and occipital lobe
Cerebral arterial circle	The Circle of Willis, the ring of arteries at the base of the brain that joins the anterior and posterior circulations
Anastomosis	A connection between two arteries that gives blood an alternate route if one path is blocked

The thirteen named arteries of the cerebral arterial supply. Compare what each one feeds.

THE THIRTEEN ARTERIES AND THEIR TERRITORIES

Artery	What it supplies
Anterior cerebral artery	The medial surface of the frontal and parietal lobes
Anterior communicating artery	A short bridge joining the right and left anterior cerebral arteries; it closes the front of the ring
Internal carotid artery	The main feed of the anterior circulation; it enters the skull and divides into the anterior and middle cerebral arteries
Middle cerebral artery	The lateral surface of the cerebrum, including the motor and sensory cortex for the body; the largest branch of the internal carotid
Posterior communicating artery	Joins the internal carotid to the posterior cerebral artery on each side, linking the anterior and posterior circulations
Posterior cerebral artery	The occipital lobe and the inferior surface of the temporal lobe
Basilar artery	Formed by the union of the two vertebral arteries; it runs up the front of the pons and supplies the brainstem
Superior cerebellar artery	A branch of the basilar artery to the superior surface of the cerebellum
Anterior inferior cerebellar artery (AICA)	A branch of the basilar artery to the anterior and inferior cerebellum
Vertebral arteries	The paired arteries that enter through the foramen magnum and unite to form the basilar artery
Posterior inferior cerebellar artery (PICA)	A branch of the vertebral artery to the inferior cerebellum and the lateral medulla
Anterior spinal artery	A single midline artery formed from both vertebral arteries; it runs down the front of the spinal cord
Posterior spinal arteries	Paired arteries from the vertebral arteries that run down the back of the spinal cord

LOCATION NAMES THE DEFICIT

Because each cerebral lobe and cortical area has its own job, the location of an injury predicts the loss. Damage to the **occipital lobe** takes vision; damage to **Broca's area** takes fluent speech while comprehension stays intact. A clinician can often estimate where a stroke struck from the pattern of symptoms alone.

The **projection fibers** of the **internal capsule** funnel nearly all traffic between the cortex and the body into one narrow band. A small stroke there severs a great deal of wiring at once, which is why such a tiny lesion can weaken an entire side of the body.

The **Circle of Willis** is the brain's insurance policy. Because the ring connects the anterior and posterior circulations, a slow blockage in one vessel can sometimes be bypassed through the communicating arteries. A sudden blockage gives the **anastomosis** no time to open, and the territory beyond it dies.